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Section 3.1: Data Storage and Backing up Data

In this section you will see what is meant by the terms '**data storage**' and '**backing up data**'.

We also discuss why it is very important that data is backed up and briefly look at some circumstances that can destroy our data.

Key Concepts of this section:

- # Know what is meant by '**Data Storage**'.
- # Understand the **difference** between **temporary storage** and **permanent storage**.
- # Be able to **discuss** why it is **important to back up data**.

Data Storage (what is it?)

Definition: "Data Storage is where we put data in a 'known' place (Save) which we can come back to later and get the data back again (Read)"



Key Words:

Data, Storage, Save/Write, Open/Read, RAM, Permanent, Volatile, Non-Volatile

Overview of Data Storage:

- # There are some terms that you need to be aware of when we talk about data storage:

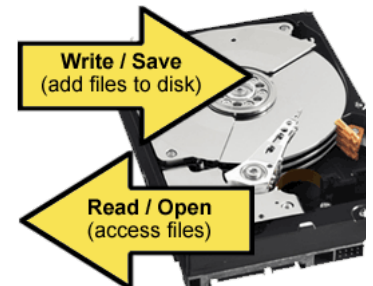
Writing of data = Storing data (also known as **saving**)
Reading data = Getting our data back (also known as **opening**)

- # Data can either be stored **temporary** or **permanent**.
 - **RAM (computer memory)** is used to store data **temporarily**
 - **Backing Storage Devices** are used to store data **permanently**.

The difference between these are explained below:

RAM (computer memory)	Backing Storage Device
Data is temporarily stored.	Data is permanently stored.
Volatile - data is erased when computer is switch off.	Non-Volatile - data is not erased when computer loses power.
Used to only store data that is currently in use . (e.g. open programs and unsaved files)	Used to store data long term . (data is only erased when we manually choose to delete it)
Very fast reading/writing of data.	Slower reading/writing of data.
	Examples of permanent storage devices include hard disk drives , DVD's and flash memory sticks .

Examples:

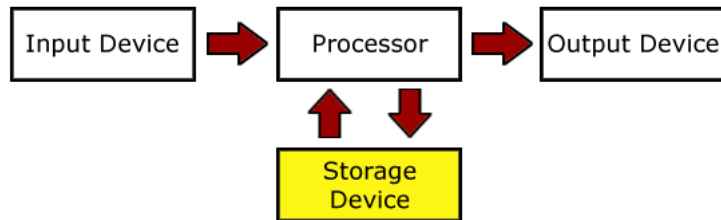


Storage devices can be saved on and read from



RAM can only store data whilst computer has power. USB sticks store data permanently.

- # The image below shows where storage devices fit into a computer system:



This diagram shows where storage devices fit into a computer system.



Without storage devices we could not run our systems, programs or save our work.

Our computers would be completely useless.

Backing up Data



Key Words:

Backup, Copy, Medium

Definition: "Backing up is the process of using a storage device to copy files and data to a different storage medium in case of a problem with the original copy"

Overview of Data Backups:

- # Backing up of data is simply the process of **making copies of your files**.
- # To back up data you need **2 pieces of equipment**:
 - **Storage Device** - The hardware that saves (writes) or reads the data from the storage medium. (DVD Player for example)
 - **Storage Medium** - The hardware that actually holds the data (Memory Sticks, Hard Disk Drives, DVD Disks etc)
- # Backing up is considered 'good practice' and helps prevent us from losing our work and files.

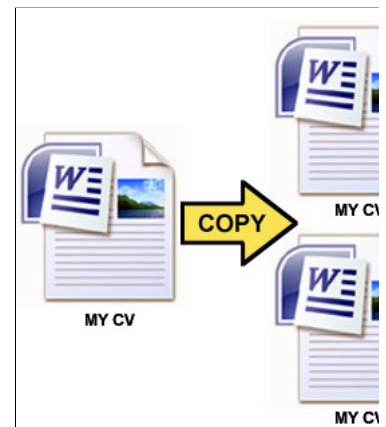
Note: There are many ways we can lose computer files. We could **accidentally delete** them, files could be **infected with a virus** or someone could even **deliberately erase** our files!

- # Backups of files should be stored in a **different location** to the original.

This is to ensure we have safe copies in the event that the originals are **destroyed by fire or floods** etc.

If the backups were in the same location as the originals then the fire would destroy the backups as well!

Examples:



Backing up data just means to 'make copies' of your files.



DVD drives are 'Storage Devices' that are used to read/write data onto DVD discs.

Why Backup Data?

- # There are many reasons why backups are made. Some of the common reasons are summarised below:

1.	Original data could be accidentally copied over or deleted .
2.	Data could be lost due to damage to the original storage device . For example: Your hard disk drive could stop spinning.
3.	You could lose your original storage device . For example: Your laptop could get stolen or you could misplace a memory stick.
4.	Hackers could access our data and delete it (or change it)
5.	Original storage device could be destroyed in a fire.
Note: Backups don't help prevent data loss in the event of viruses. if the original file had a virus BEFORE the it was backed up then the copy will also be infected.	



DVD discs are example of 'Storage Medium' and hold the data.



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